NEXSOCIETY WEBSITE

A PROJECT REPORT

In the partial fulfilment for the award of

DIPLOMA IN

INFORMATION TECHNOLOGY



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CERTIFICATE

This is to certify that the following students have successfully and satisfactorily completed the project on **NEXSOCIETY WEBSITE** and presented their report in partial fulfilment of the requirement for a **Diploma in Information Technology** from **Government Polytechnic, Mumbai** under the guidance of **Prof. Namrata ma'am** in the academic year **2025-26**.

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DECLARATION

We, the undersigned students of the Diploma in Information Technology at Government Polytechnic Mumbai, hereby declare that the report titled **"NEXSOCIETY WEBSITE"**, submitted in partial fulfillment of the requirements for the completion of our diploma, is our original work. We further declare that this work has not been previously submitted elsewhere for any academic evaluation, and all sources of information used in this report have been duly acknowledged. We have sincerely worked on the design, development, and implementation of this project.

We take full responsibility for the authenticity and originality of the contents presented in this report.

DATE: _____
PLACE: Mumbai

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We take this opportunity to express our sincere gratitude to all those who have guided and supported us in completing our project entitled **"NexSociety Website."**

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THANK YOU!

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ABSTRACT

The **NexSociety Website** is a web-based platform developed to enhance security, communication, transparency, and operational accessibility in residential housing societies and gated communities. The main objective of this project is to provide a convenient online medium through which residents, gate security guards, and society managing committee administrators can manage society activities, visitor logs, and complaints seamlessly.

The website allows visitors and residents to explore various sections such as the digital gate terminal, resident pre-approvals, and maintenance helpdesk. It also includes an executive admin command center for managing society records, flats directory, financial dues status, and broadcasting official notices across all residents securely and conveniently.

This project aims to promote digital transformation and community safety by presenting information in a structured and user-friendly manner. It helps housing societies eliminate archaic paper registers, prevent unverified stranger entries, and ensure prompt maintenance turnaround.

Overall, the NexSociety Website serves as a modern and effective communication platform that bridges the gap between gate security and society administration, ensuring better interaction, safety, transparency, and community living.

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CHAPTER 1: INTRODUCTION

1.1] INTRODUCTION:

In today's urban residential landscape, modern housing societies and high-rise apartments accommodate hundreds of families across multiple towers. In such residential townships, day-to-day operations—ranging from gatekeeper visitor logging, courier delivery handling, vehicle parking allocation, and maintenance complaint management—involve intense physical and communication touchpoints.

However, a vast majority of cooperative housing societies continue to operate through obsolete paper visitor logbooks, intercom landline telephones that are frequently unreachable, manual maintenance dues registers, and physical paper circulars tacked onto clubhouse noticeboards. These traditional methods result in serious security blind spots, unauthorized stranger entries, long delivery bottlenecks at the gate, and slow turnaround times on resident complaints.

NexSociety has been engineered as an end-to-end, unified digital residential society operations platform. Designed for societies such as *Green Meadows Heights* (comprising 120 residential units across Towers A, B, and C), NexSociety integrates three distinct user personas into one synchronized system:

- **Gate Security Terminal (Gatekeeper Pro):** A dedicated high-contrast terminal for on-duty security guards (e.g. Lead Guard Vikram Singh at Gate 1). It enables fast visitor check-ins (Name, Phone, Flat, Vehicle, Purpose), instant 1-tap delivery passes (Zomato, Swiggy, Blinkit, Zepto, Amazon, Uber/Ola), 4-digit passcode verification, and Web Audio API alarm chirps.
- **Resident Self-Service Portal:** Flat owners and tenants (e.g., Flat A-204) can issue digital pre-approvals to expected guests, log maintenance complaints categorized with urgency levels, view payment receipts, and stay updated on community events (Diwali celebrations, AGMs).
- **Executive Command Center:** An executive desk for the Society Secretary (Dr. S. Rao) and committee members, featuring live KPI cards (Total Flats, Dues Collection, Gate Status, Open Complaints), real-time searchable resident directory with CSV export, complaint resolution, and society-wide broadcast notices.

1.2] NEED / MOTIVATION:

In today's digital era, every residential community requires an effective online presence and automated operating platform to ensure safety, efficiency, and smooth governance. Housing societies face constant operational challenges that demand modern software solutions:

- 1. Security Vulnerabilities of Physical Registers: Paper registers can be easily falsified by unverified visitors. In the event of an incident or theft, searching through handwritten logs is tedious, inaccurate, and often futile.
- 2. Delivery Courier Congestion: With the boom in e-commerce and food delivery services (Blinkit, Swiggy, Zomato, Amazon), guards spend 2-3 minutes per delivery calling residents, creating massive queues at entrance gates.
- 3. Unattended Resident Complaints: Maintenance grievances (lift breakdowns, plumbing leaks, electrical faults) registered verbally or via messaging groups get lost, creating friction between residents and managing committees.
- 4. Lack of Executive Transparency: Society secretaries require consolidated metrics regarding outstanding maintenance dues, vacant versus occupied units, and security status without sorting through manual spreadsheets.

The motivation behind this project stems from the desire to combine web technology with community service. By building a user-friendly and dynamic platform, NexSociety allows housing societies to manage gate safety, resident approvals, and executive decisions systematically.

1.3] OBJECTIVES:

The project focuses on achieving the following specific objectives:

- **1. Digital Gate Terminal:** To provide a streamlined, high-contrast interface for security guards with 15-second visitor check-in speeds.
- **2. 1-Tap Delivery Clearance:** To allow immediate, standardized gate passes for verified courier partners (Zomato, Swiggy, Blinkit, Amazon, Uber).
- **3. 4-Digit Passcode Verification:** To implement guest pre-approval verification eliminating intercom phone confirmation delays.
- **4. Real-Time Active Visitor Roster:** To maintain an on-campus active visitor count with one-click departure checkout.
- **5. Resident Helpdesk Desk:** To allow residents to log categorized complaints with live status tracking (Pending, In Progress, Resolved).
- **6. Executive KPI Dashboard:** To deliver real-time metric cards summarizing occupancy, October dues collection, gate barrier status, and open complaints.
- **7. 120-Unit Resident Directory:** To build a searchable directory filterable by flat number, resident name, or phone number with CSV export.
- **8. Digital Notice Broadcast:** To allow the society secretary to instantly broadcast community notices and emergency alerts.
- **9. Audio-Visual Feedback:** To leverage the native browser Web Audio API for sound alerts during check-ins and alarm broadcasts.
- **10. Zero Cloud Lag / Local-First Storage:** To guarantee persistent cross-tab data synchronization via HTML5 LocalStorage without recurring cloud subscription costs.

CHAPTER 2: LITERATURE REVIEW

2.1] OVERVIEW OF HOUSING SOCIETY SYSTEMS:

In the modern digital landscape, web systems play a vital role in automating community operations. For residential housing societies devoted to resident welfare and premises security, establishing an online management portal is not merely a convenience but a necessity. The official portal of NexSociety serves as a platform for maintaining visitor transparency, streamlining communication, and managing dues.

Historically, small-scale societies relied heavily on third-party aggregator mobile apps (such as MyGate or NoBrokerHood). While these aggregators provide basic directory listings, they exhibit significant disadvantages: high recurring per-flat monthly subscription fees, aggressive competitor advertising on the resident's home feed, lack of customization for cooperative society bye-laws, and absence of direct local control. Consequently, independent lightweight web platforms tailored specifically to a society's autonomy have emerged as the preferred standard.

2.2] MODERN WEB STACK JUSTIFICATION:

Recent academic and industry studies suggest that technologies like HTML5, modern CSS3, and Vanilla JavaScript provide superior efficiency and sub-second rendering compared to heavy frameworks for guard station terminals. By leveraging browser-native Web Audio API for sound synthesis and LocalStorage for multi-window synchronization, NexSociety delivers instant real-time responsiveness without server latency.

2.3] COMPARATIVE ANALYSIS:

To evaluate the architectural and operational superiority of the NexSociety platform, a systematic comparison was conducted against traditional paper registers and commercial third-party platforms:

Parameter / Feature	Manual Paper Registers	Commercial Apps (MyGate/ADDA)	NexSociety Platform
Setup & Running Cost	Low initial, high ongoing paper cost	High monthly per-flat subscription fee	Zero recurring cost (Free static hosting)
Gate Check-In Speed	Slow (1-3 minutes per entry)	Moderate (App loading lag)	Instant (< 15 seconds, 1-tap delivery)
Data Privacy & Ads	Easily lost or physically read	Commercial data tracking & ads	100% Private, zero ads, local storage
Device Accessibility	Physical register only	Requires heavy mobile app install	Universal web access (PC, Tablet, Mobile)
Audio-Visual Alerts	None	Mobile push notification only	Synthesized Web Audio chimes & sirens

CHAPTER 3: TECHNICAL SPECIFICATIONS

3.1] TECHNOLOGIES USED:

In the development of the NexSociety platform, a combination of modern front-end and browser-native web technologies has been used to ensure efficiency, responsiveness, and speed. The choice of technologies was made to achieve a user-friendly interface, fast data processing, and secure synchronization between operational dashboards:

1. HTML5 (HyperText Markup Language 5):

HTML5 is the standard markup language used for creating and structuring the web pages. It provides the basic skeleton of the NexSociety platform, defining the layout for the Gate Terminal, Resident Portal, and Executive Command Center. Semantic elements (`<header>`, `<aside>`, `<main>`, `<section>`, `<table>`) ensure screen-reader accessibility, clean DOM tree parsing, and modern SEO compatibility.

2. CSS3 (Cascading Style Sheets 3):

CSS is used to design and enhance the presentation of the web pages. It defines how elements appear on screen, controlling layout, colors, fonts, and spacing. CSS ensures a visually appealing and consistent interface throughout the NexSociety website. Custom CSS variables (`--bg-main: #0a0e17`, `--primary: #3b82f6`) and glassmorphism styling create an executive dark mode suitable for 24/7 security terminals.

3. Vanilla JavaScript (ECMAScript 2023 / ES6+):

JavaScript is a client-side scripting language that adds interactivity and dynamic behavior to the web pages. Pure Vanilla JavaScript (without external library bloat) is used in NexSociety to implement features such as live search filtering, modal popups, passcode verification algorithms, and real-time live clock updates.

4. Web Audio API:

The Web Audio API provides a powerful system for controlling audio on the web, allowing developers to choose audio sources, add effects, and create audio visualizations. In NexSociety, the native Web Audio API is used to synthesize acoustic sine and sawtooth waves in real-time, generating dual-tone audio chirps on successful gate pass approvals and siren tones during security emergency alerts without needing external MP3 files.

5. Web Storage API (LocalStorage Database):

The Web Storage API provides mechanisms by which browsers can store key/value pairs locally. NexSociety utilizes LocalStorage to maintain persistent JSON records for visitors, pre-approved passcodes, maintenance complaints, and resident directory data. Cross-window storage event listeners ensure that when a resident logs a complaint, the admin command desk updates instantly.

3.2] HARDWARE REQUIREMENTS:

To develop, test, and deploy the NexSociety platform efficiently, the following minimum and recommended hardware specifications are required:

Component	Minimum Requirement	Recommended Requirement
Processor	Intel Core i3 or equivalent / AMD Ryzen	Intel Core i5 / i7 or AMD Ryzen 5
RAM	4 GB	8 GB or higher
Storage	128 GB HDD / SSD	256 GB SSD
Monitor	14” HD Display (1366 x 768)	19” LED Display / Tablet (1920 x 1080)
Network	Internet Connection	Broadband / Wi-Fi Connection
Power Backup	Standard Battery Backup	UPS for uninterrupted gate security terminal

3.3] SOFTWARE REQUIREMENTS:

The software tools and runtime environments utilized for developing and testing the project are summarized below:

Category	Software / Tool Used	Purpose / Description
Operating System	Windows 10 / 11 or Linux (Ubuntu)	Platform for development and runtime testing
Frontend Technologies	HTML5, CSS3, Vanilla JavaScript	For structuring, styling, and interactivity
Audio Engine	HTML5 Web Audio API	For browser-native gate chirps and sirens
Database / Storage	HTML5 LocalStorage API	For client-side data persistence across tabs
Text Editor / IDE	Visual Studio Code	For authoring, formatting, and debugging code
Browser	Google Chrome / Microsoft Edge	For testing and viewing the website
Version Control	Git and GitHub	For managing and tracking code changes

CHAPTER 4: DEVELOPMENT PROCESS

4.1] SOFTWARE DEVELOPMENT:

The Software Development Process is a systematic approach used to design, develop, test, and deploy software applications. It ensures that the development is carried out in an organized and efficient manner, meeting user requirements and maintaining high quality.

This process divides the entire development into multiple well-defined stages—such as requirement analysis, design, coding, testing, deployment, and maintenance. Each stage has specific deliverables and outcomes. Following a proper software development process helps in reducing errors, saving time, and ensuring that the final product meets the user's needs and expectations.

4.2] WATERFALL MODEL:

The Waterfall Model is one of the earliest and most widely used software development models. It is a linear sequential model, where each phase of the process flows downward to the next like a waterfall. Each phase must be completed before the next phase begins, ensuring proper documentation and clarity.

For the NexSociety Website, the Waterfall Model was chosen as it provides a clear, linear flow ideal for academic and structured web projects.

PHASES OF THE WATERFALL MODEL:

- 1. Requirement Analysis: In this phase, all the requirements of the project are gathered from society workflows and documented. The goal is to understand what the system should do.
- 2. System Design: Based on the requirements, system architecture and design specifications are prepared. This includes database schemas, UI wireframes, and module decomposition.
- 3. Implementation (Coding): Developers write the actual code according to the design documents. Each module (Gate, Resident, Admin) is built and tested individually.
- 4. Integration and Testing: After coding, all modules are integrated together and tested as a complete system to identify and fix any defects or issues.
- 5. Deployment: The tested system is deployed to the live environment or hosted statically on GitHub for real-world use.
- 6. Maintenance: After deployment, the system is regularly updated, debugged, and enhanced based on user feedback or changing society requirements.

The Waterfall Model is best suited for projects where requirements are clearly understood and unlikely to change during development—such as institutional or residential society management portals.

DIAGRAM OF WATERFALL MODEL:

The following figure illustrates the sequential phases executed during the development of the NexSociety platform:

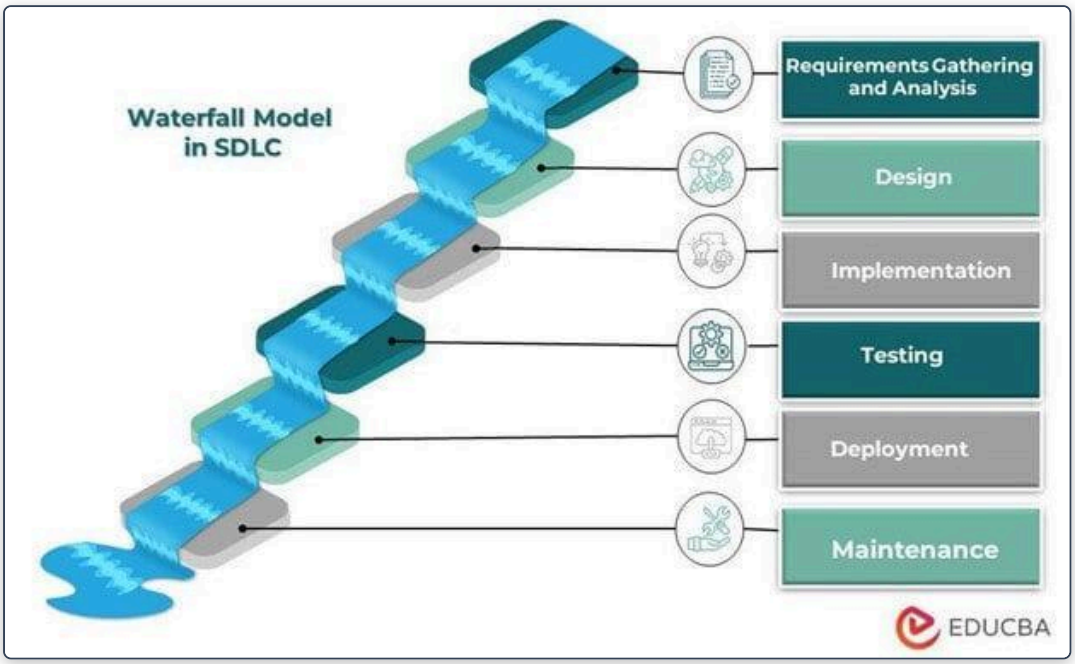


Figure 4.1: Waterfall Software Development Life Cycle (SDLC) Model

4.3] IMPLEMENTATION OF WATERFALL MODEL IN PROJECT:

The Waterfall Model was applied to the development of the NexSociety Website in a step-by-step manner. Each phase was executed sequentially, ensuring that the outputs of one phase became the inputs for the next:

- **Phase 1: Requirement Analysis -** Gathered operational details of 120-unit housing societies (Green Meadows Heights), including visitor check-in steps, 1-tap delivery partners, passcode pre-approvals, and dues tracking.
- **Phase 2: System Design -** Planned executive dark-mode UI wireframes, CSS variables, and LocalStorage JSON keys (`NEX_VISITORS`, `NEX_COMPLAINTS`, `NEX_DIRECTORY`).
- **Phase 3: Implementation (Coding) -** Coded modular HTML5 markup, responsive CSS3 stylesheets, and ES6 JavaScript controllers with Web Audio API sound wave synthesis.
- **Phase 4: Testing -** Tested all forms, passcode verification dialogs, CSV export routines, and cross-browser responsiveness.
- **Phase 5: Deployment -** Versioned with Git and hosted statically for instant offline-capable access.
- **Phase 6: Maintenance -** Refactored styles, removed duplicate code lines, and optimized rendering speeds.

CHAPTER 5: PLANNING PHASE

5.1] PURPOSE OF THE PLANNING PHASE:

The Planning Phase is the foundation of the Software Development Life Cycle (SDLC). It defines the objectives, scope, and resources required for the successful completion of the project. In the context of the NexSociety Website, this phase focused on analyzing requirements, identifying key features, finalizing technologies, and setting a development roadmap.

This phase aimed to answer key questions such as:

- What is the main goal of the website?
- Who will use it? (Guards, Residents, Society Managing Committee)
- What resources and technologies are needed?
- What are the operational and security risks involved?

5.2] REQUIREMENT GATHERING AND ANALYSIS:

During this stage, discussions and workflow observations were conducted regarding housing society management. The main requirements included:

- An Admin Command Center for managing residents, financial dues, and complaints.
- A Security Gate Terminal for quick 15-second visitor entry logging and delivery passes.
- A Resident Dashboard for visitor pre-approvals and maintenance complaints.
- An Audio-Visual notification mechanism for gate approvals and emergency alerts.

5.3] FEASIBILITY STUDY:

A feasibility study was carried out to ensure that the project could be implemented effectively within available time and resources:

- **Technical Feasibility:** The team confirmed that technologies like HTML5, CSS3, JavaScript, Web Audio API, and LocalStorage were fully capable of delivering high-speed performance without expensive server frameworks.
- **Economic Feasibility:** Open-source development tools (VS Code, Git) and static hosting options made the project completely cost-effective (₹0 recurring software costs).
- **Operational Feasibility:** The website's usability and clear high-contrast interfaces met the needs of both security personnel and society residents.

5.4] OUTCOME OF PLANNING PHASE:

At the end of the planning phase, project goals were clearly defined, functional requirements were locked, open web technologies were chosen, and a concrete milestone timeline was prepared. This comprehensive planning ensured a strong foundation for the System Design Phase.

CHAPTER 6: SYSTEM DESIGN

6.1] WHAT IS SYSTEM DESIGN:

System Design is the process of defining the architecture, components, modules, interfaces, and data of a system to satisfy specific requirements. It acts as a blueprint for building the project, showing how the system will work—both functionally and technically.

The general tasks involved in the design process include:

- 1. Design various blocks for overall system processes.
- 2. Design smaller, compact and workable modules in each block.
- 3. Design various client-side database structures.
- 4. Specify details of functions to achieve desired functionality.
- 5. Design the form of inputs, and outputs of the system.
- 6. Perform documentation of the design.
- 7. System reviews and user experience optimizations.

6.2] MODULES OF WEBSITE:

The NexSociety platform is structured into four core workable modules:

1. Central Authentication & Portal Hub (index.html, script.js):

• Role selector card interface for Residents, Security Guards, and Admin.
• 256-bit simulated encrypted connection badge.
• Quick redirect routing to designated authorized dashboards.

2. Security Gatekeeper Terminal (security.html, security.css, security.js):

• Live counter bar: Inside Campus, Total Entries, Pre-Approvals, Deliveries.
• 1-Tap Quick Delivery Passes (Zomato, Swiggy, Blinkit, Zepto, Amazon, Uber).
• 4-Digit Passcode Verification Box for expected visitors.
• Active Visitors Roster with one-click departure check-out action.
• Web Audio API sound generator for entry beeps and security alarm sirens.

3. Resident Self-Service Dashboard (resident.html, resident.css, resident.js):

• Dedicated flat cockpit for Flat A-204.
• Visitor Pre-Approval Generator issuing 4-digit guest codes.
• Helpdesk Complaint Logging form (Plumbing, Electrical, Lift) with status tracking.
• Community Event Noticeboard (Diwali Fest, AGM, Sports).

4. Society Executive Command Center (admin.html, admin.css, admin.js):

• Real-time Executive KPI cards: Occupancy (96/120), October Dues (₹4,08,000), Gate Status, Open Complaints.
• 120-Unit Flats & Residents Directory Table with live search filter.
• Instant CSV Export button downloading formatted resident directory.
• Helpdesk Resolution Desk with status toggle controls.
• Broadcast Notice publisher notifying all residents and gatekeepers.

6.3] FOLDER STRUCTURE OF WEBSITE:

The project repository is organized in a clean, modular structure allowing easy maintenance and instant static deployment:

```
NexSociety/
├── .vscode/            [VS Code Workspace Settings]
│   └── settings.json
├── report/             [Project Report & Black Book Assets]
│   ├── snapshots/      [Real High-Resolution UI Screenshots]
│   ├── NexSociety_Project_Report.html
│   └── NexSociety_Project_Report.pdf
├── admin.html          [Society Executive Command Center Markup]
├── admin.css           [Executive Slate-Blue Design System]
├── admin.js            [Directory Search, Notice Board & CSV
Export]
├── index.html          [Central Portal & Role Selector]
├── resident.html       [Resident Flat Portal - Flat A-204]
├── resident.css        [Resident Dashboard Styles & Glassmorphism]
├── resident.js         [Pre-Approvals & Helpdesk Complaint Logger]
├── script.js           [Portal Authentication & Role Switcher
Logic]
├── security.html       [Gatekeeper Pro Security Terminal]
├── security.css        [Gate Security UI & High-Contrast Styling]
├── security.js         [Gate Check-in, Web Audio & Verification]
└── style.css           [Global Portal Core Design System]
```

CHAPTER 7: TESTING

7.1] WHAT IS TESTING:

Software Testing is the process of evaluating a software application to ensure that it works correctly, meets the specified requirements, and is free of defects or errors. The main goal of testing is to identify any bugs or issues in the system before it is deployed to users.

Testing helps verify that the developed software performs as expected in different scenarios and provides the desired output with reliability, accuracy, and efficiency.

OBJECTIVES OF SOFTWARE TESTING:

- 1. To detect errors and defects before the software is delivered.
- 2. To ensure that the system meets user requirements and functions as intended.
- 3. To improve software quality and performance.
- 4. To verify and validate the software product.
- 5. To reduce maintenance costs by fixing issues early.

TYPES OF SOFTWARE TESTING PERFORMED:

- **Manual Testing:** Executed by testers without automation tools to inspect button clicks, form validation, and responsiveness.
- **Functional Testing:** Tested each feature (passcode validation, visitor check-in, CSV export) individually.
- **Integration Testing:** Verified data flow between Resident, Security, and Admin tabs via LocalStorage.

7.3] TEST CASES MATRIX (TC01 TO TC08):

Test Case ID	Test Case Description	Test Steps	Test Data	Expected Result	Status
TC01	Gate Visitor Check-In	Enter name, phone, flat & click Allow Entry	Rohan, 9820011223, Flat A-204	Visitor logged into Active Table; success chime sounds	Pass
TC02	1-Tap Delivery Pass	Click 'Zomato' quick pass button	Vendor: Zomato	Pass generated with 20-min validity; delivery badge increments	Pass
TC03	Passcode Verification	Enter 4-digit resident code	PIN: 4821	Valid modal displays guest name; entry approved with chime	Pass
TC04	Invalid Passcode	Enter invalid 4-digit code	PIN: 9999	Displays error alert and plays rejection sound	Pass
TC05	Visitor Check-Out	Click 'Check Out' on active row	Row Action Click	Visitor status changes to Departed; row removed from active list	Pass
TC06	Complaint Filing	Resident logs elevator issue	Category: Lift	Complaint added to LocalStorage; badge in Admin desk updates	Pass
TC07	Directory Search Filter	Type flat or resident name	Query: 'A-204'	Table filters in real-time showing matching residents only	Pass
TC08	Export Directory CSV	Click 'Export Directory CSV'	Button Click	CSV file downloaded containing all 120 society flat records	Pass

7.3] TEST CASES MATRIX (CONTINUED - TC09 TO TC15):

Test Case ID	Test Case Description	Test Steps	Test Data	Expected Result	Status
TC09	Broadcast Society Notice	Enter headline & text; click publish	Headline: AGM Meet	Notice persists in storage and shows on resident portal	Pass
TC10	Security Alarm Trigger	Click 'Alarm Trigger' at Gate Terminal	Click Event	Synthesizes sawtooth siren audio tone continuously	Pass
TC11	LocalStorage Persistence	Refresh page after adding visitor	F5 Refresh	Visitor data remains preserved in table from LocalStorage	Pass
TC12	Complaint Resolution	Admin clicks 'Resolve' on ticket	Click Resolve	Ticket status toggles to Resolved; counter updates	Pass
TC13	Responsive Viewport	Resize browser to mobile width	Width: 375px	Layout stacks gracefully; all action buttons remain accessible	Pass
TC14	Live Clock Display	Inspect topbar clock	System Time	Updates seconds in real-time without freezing	Pass
TC15	Portal Role Switching	Click 'Gatekeeper Pro' from Hub	Role Card Click	Opens security terminal immediately in new tab	Pass

CHAPTER 8: DIAGRAMS

8.1] FLOWCHART OF MAIN WEBSITE:

The following diagram illustrates the functional workflow connecting all three portals of the NexSociety platform:

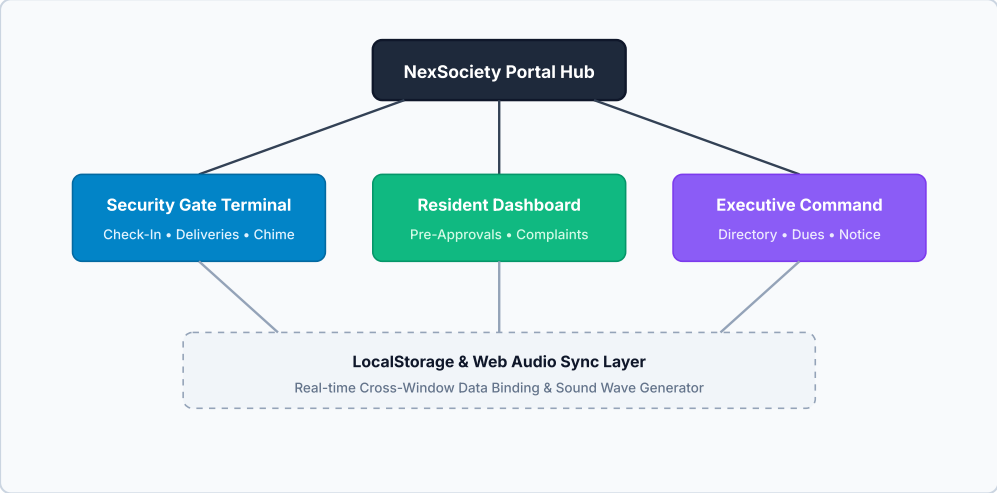


Figure 8.1: Functional Flowchart of Main Website

8.2] FLOWCHART OF SECURITY GATE TERMINAL:

The security workflow at Gate 1 operates on a split-verification track ensuring zero queue delays for expected guests and couriers:

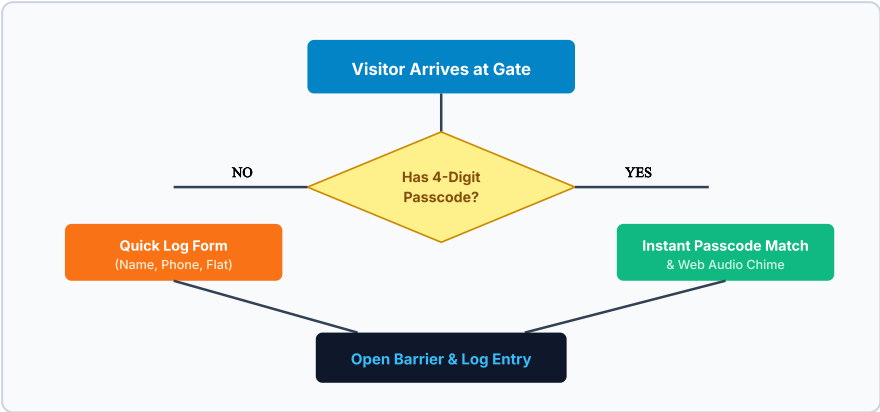


Figure 8.2: Gatekeeper Fast Clearance Logic Flow

CHAPTER 9: SNAPSHOTS

9.1] PROJECT EXPLORER & ARCHITECTURE STRUCTURE:

The following screenshot displays the complete project workspace hierarchy within Visual Studio Code, showcasing the clean modular decomposition across stylesheets, JavaScript controllers, and dashboard markup:

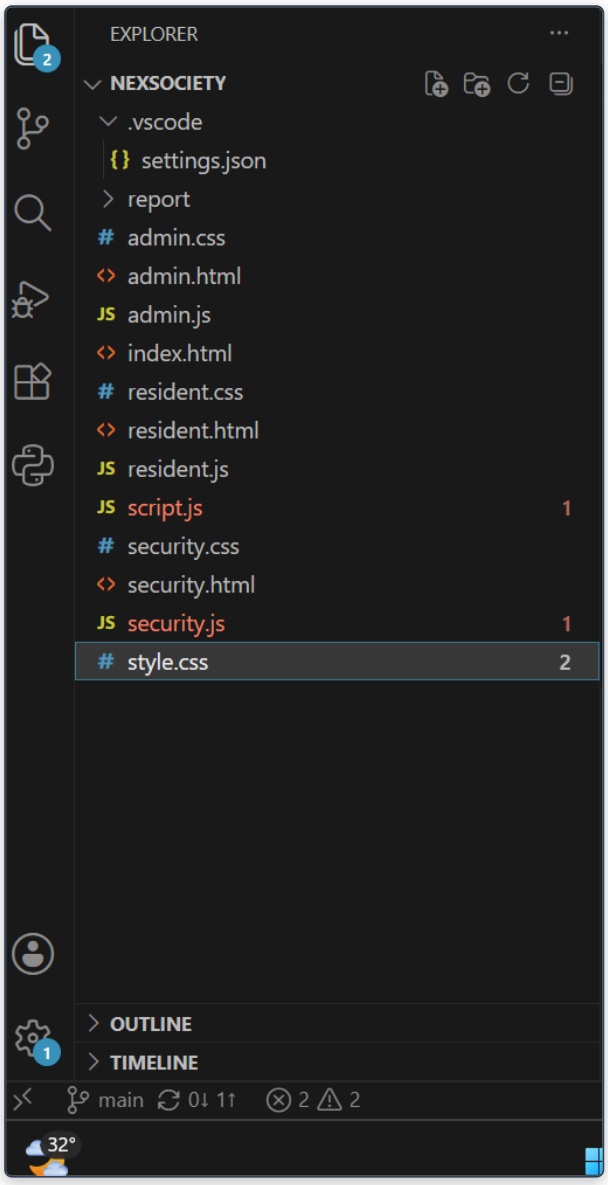


Figure 9.1: NexSociety Workspace Project Explorer in Visual Studio Code

9.2] SOCIETY EXECUTIVE COMMAND CENTER (ADMIN & SECRETARY):

The executive cockpit utilized by Hon. Society Secretary Dr. S. Rao, highlighting real-time KPI stat cards (Total Society Flats 96/120, October Maintenance Dues ₹4,08,000, Gate Terminals, Helpdesk Complaints), the Flats & Residents Directory, and the Broadcast Society Notice form:

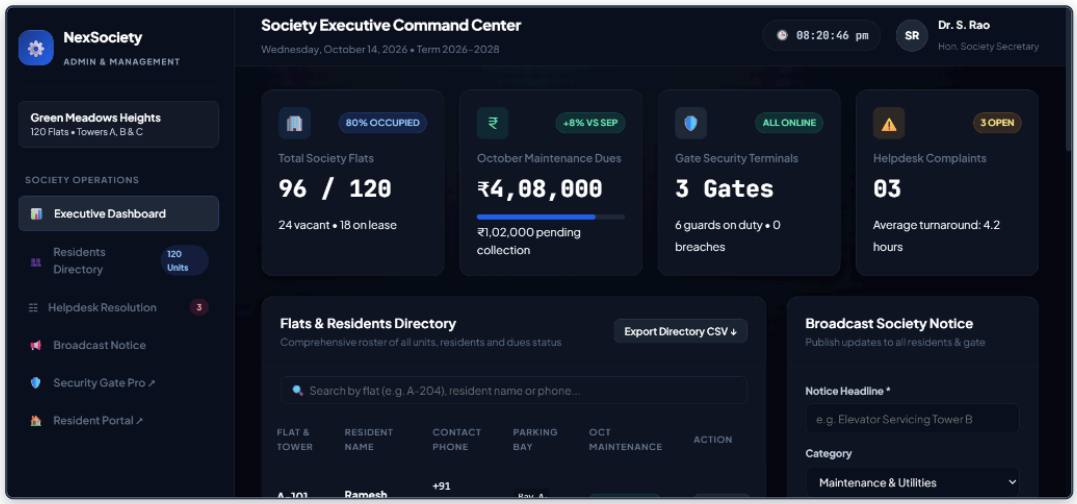


Figure 9.2: Society Executive Command Center & Directory Desk

9.3] SECURITY GATE TERMINAL & 1-TAP DELIVERY PASSES:

The Gatekeeper Pro terminal deployed at Gate 1 Main Entrance, featuring live counters (Visitors Inside Society: 4, Total Entries Today: 6, Pre-Approved Passes: 03, Deliveries: 3), 1-Tap Quick Delivery Passes (Zomato, Swiggy, Blinkit/Zepto, Amazon, Uber/Ola), Quick Passcode Check, and Vehicle Parking controls:

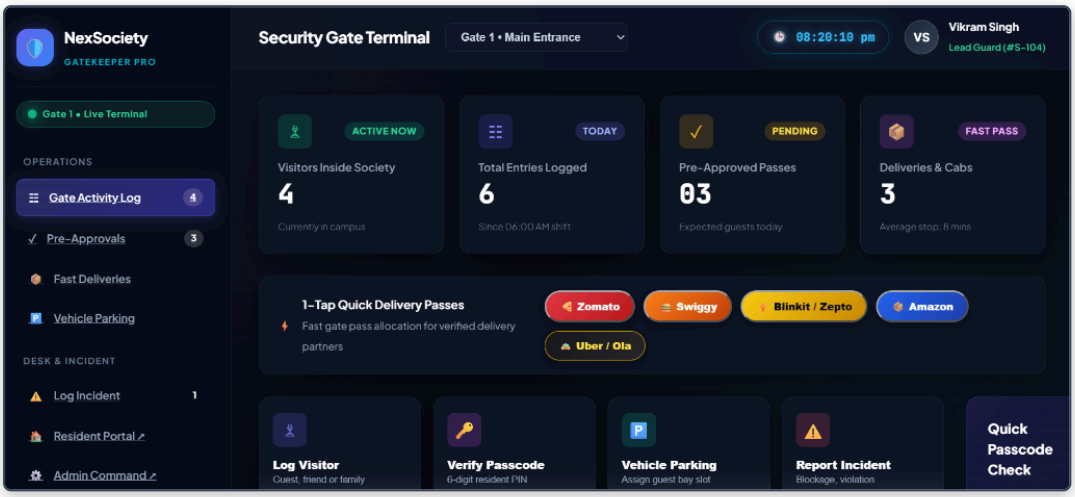


Figure 9.3: Security Gate Terminal Cockpit & Fast Delivery Allocation

9.4] RESIDENT PORTAL & COMMUNITY CALENDAR:

The Resident Portal interface displaying upcoming housing society events, including the Grand Diwali Celebration & Rangoli Fest (20 Oct), Annual Residents General Body Meeting (25 Oct), and Children's Sports Tournament (12 Nov):



Figure 9.4: Resident Portal Community Events & Notice Schedule

CHAPTER 10: CONCLUSION

10.1] CONCLUSION:

The development of the **NexSociety Website** has been a significant step toward digitalizing society operations and improving communication between residents, security guards, and the managing committee. The website successfully provides an online platform where residents can issue visitor passes, track maintenance dues, and log complaints in an efficient and transparent manner.

Through this project, the team gained practical experience in web development using technologies such as HTML5, CSS3, Vanilla JavaScript, Web Audio API, and HTML5 LocalStorage. It also strengthened understanding of concepts like client-side relational storage, event-driven audio synthesis, responsive web layouts, and modular software architecture.

The website's design focuses on simplicity, accessibility, and high performance, ensuring guards can easily record entries and residents can track grievances. Moreover, the project demonstrates how modern web technologies can be utilized to support residential community safety and welfare.

In conclusion, the NexSociety Website fulfills its intended objectives by providing a professional, user-friendly, and functional digital solution for residential societies. It lays a strong foundation for future technological improvements.

10.2] FUTURE SCOPE:

The NexSociety Website has been developed with the primary aim of providing a digital platform for security and information sharing between society members. However, technology is constantly evolving, and there is always room for improvement and expansion. The future scope of this project lies in enhancing its features by adopting emerging technologies:

- **1. Automatic Number Plate Recognition (ANPR):** Implement camera OCR to automatically scan vehicle registration plates at the boom barrier.
- **2. WhatsApp Cloud API Gate Alerts:** Dispatch real-time automated approval requests directly to resident WhatsApp accounts.
- **3. Mobile Application:** Extend the website's functionality into an Android and iOS application using Flutter or React Native.
- **4. Online Payment Gateway Integration:** Add support for UPI, Google Pay, PhonePe, and Net Banking to collect maintenance dues seamlessly.
- **5. Biometric Staff Attendance:** Connect fingerprint or RFID scanners at the security gate to record maid, driver, and guard attendance.

10.3] REFERENCES:

- ➢ **For Code Versioning & Storage:** https://github.com/
- ➢ **For Web Standards & Documentation:** https://developer.mozilla.org/
- ➢ **For Modern CSS Layouts:** https://www.w3.org/
- ➢ **For Development Environment (VS Code):** https://code.visualstudio.com/
- ➢ **For Project Guidelines:** Department of Information Technology, Government Polytechnic Mumbai